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FLEXIBLE CIRCUITS CONFIGURED FOR HEAT SPREADING IN A WEARABLE DISPLAY

Abstract

A head-worn display (HWD) configured to provide an extended reality (XR) image to a user include includes a flexible circuitry configured to electrically couple a component of the HWD to a processor or power source. Additionally, this flexible circuit includes a heat tab configured to thermally couple the flexible circuit to the component of the HWD. Due to the flexible circuitry being thermally coupled to the component of the HWD, heat generated by the component of the HWD is spread across the surfaces of the flexible circuitry, lowering the temperature of the component.

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Background/Summary

BACKGROUND

[0001] Head-worn displays (HWDs) are commonly configured to direct light representative of an image to the eye of a user such that the image is viewable to the user in a real-world space. To this end, some HWDs include a projector that includes one or more light-emitting components each configured to emit light, such as laser light, representative of at least a portion of an image toward a lightguide of the HWD. For example, the projector emits the light such that the light is first received at an incoupler of the lightguide. This incoupler then directs the light into the main body of the lightguide such that the light propagates through the body of the lightguide toward an outcoupler of the lightguide. The outcoupler then directs the received light out of the lightguide and toward the eyes of a user. The light directed by the outcoupler then forms an exit pupil near the eyes of the user, allowing the user to view the image represented by the light in a real-world space.

[0002] However, while the projectors of these HWDs are operating, the projectors generate heat based on the amount of power that the projectors consume. The heat generated by a projector, in turn, heats the portions of the HWD that encase the projector, such as a frame of the HWD. Heating these portions of the frame of the HWD in this way increases the likelihood that a hotspot forms on the frame of the HWD, making use of the HWD uncomfortable for the user and negatively impacting user experience.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The present disclosure may be better understood, and its numerous features and advantages made apparent to those skilled in the art by referencing the accompanying drawings. The use of the same reference symbols in different drawings indicates similar or identical items.

[0004] FIG. 1 is a diagram of an example display system housing a projection system and configured to spread heat using flexible circuits, in accordance with some embodiments.

[0005] FIG. 2 is a diagram illustrating a projection system that projects images directly onto the eye of a user, in accordance with some embodiments.

[0006] FIG. 3 is a block diagram of an example display having one or more heat-generating components, in accordance with some embodiments.

[0007] FIG. 4 is a diagram of an example flexible circuit, in accordance with some embodiments.

[0008] FIG. 5 is a diagram of an example flexible circuit configured for heat spreading, in accordance with embodiments.

[0009] FIG. 6 is a diagram of a display system including a flexible circuit configured for heat spreading, in accordance with some embodiments.

[0010] FIG. 7 is a partially transparent view of a head-worn display (HWD) including flexible circuits configured for heat spreading, in accordance with some embodiments.

DETAILED DESCRIPTION

[0011] Systems and techniques herein are directed to head-worn displays (HWDs) (e.g., extended reality HWDs) configured to direct light toward the eyes of a user such that one or more extended reality (XR) images are presented to the user. For example, an HWD has a housing (e.g., frame) resembling eyeglasses and includes one or more lenses containing a lightguide to direct light representative of an image to the eye of the user. Herein, the combination of the lens and lightguide is referred to as an “optical combiner,” “optical combiner lens,” or both. Such a lightguide, for example, includes one or more incouplers, exit pupil expanders (EPEs), and outcouplers configured to direct light representing an XR image from a projector to the eye of the user. As an example, the lightguide includes an incoupler configured to receive light representing an XR image emitted from a projector of a display circuitry and direct the received light into the lightguide such that the light propagates through the lightguide using total internal reflection (TIR), partial internal reflection

(PIR), or both. The light then propagates through the lightguide until the light is received at an outcoupler of the lightguide. In response to receiving the light, the outcoupler directs the light out of the lightguide and towards the eye of the user such that the light forms an exit pupil representative of the XR image near the eye of the user. This exit pupil, for example, represents the location along the optical path where the beams of the light, as directed by the lightguide, intersect. Further, some lightguides include an exit pupil expander (EPE) configured to receive light from the incoupler and direct light towards the outcoupler. Such an EPE, for example, is further configured to direct the light toward the outcoupler such that the size of the exit pupil is increased (e.g., the exit pupil is expanded).

[0012] Further, to help improve user experience, some HWDs include additional circuitry that expands the functionality of the HWD. That is to say, in addition to a display circuitry to provide an XR image to a user, some HWDs include other circuitry that provides additional functionality. For example, some HWDs include a connectivity circuitry configured to communicatively couple the HWD to one or more other devices (e.g., smartphones, computers, tablets), a network (e.g., local area network, wide area network), or both using one or more wireless communication protocols (e.g., Bluetooth, Wi-Fi, cellular protocols). As another example, some HWDs include sensing circuitry configured to take one or more measurements of the user, the HWD, or both. The HWD then uses these measurements to determine the gaze of the user, focus of the user, objects in front of the user, a pose of the user, a location of the user, or any combination thereof. To this end, the sensing circuitry includes one or more cameras, infrared sensors, accelerometers, radar sensors, proximity sensors, and the like. As yet another example, some HWDs include audio circuitry that includes one or more speakers, digital signal processors, microphones, and the like configured to receive and output audio (e.g., music, sounds, voice) for one or more streaming applications, music applications, video applications, audio calling applications, video calling applications, and the like running on the HWD or a device communicatively coupled to the HWD.

[0013] Additionally, the HWD includes a processor configured to manage the functionalities of the circuitries (e.g., display circuitry, connectivity circuitry, sensing circuitry, audio circuitry), execute applications, manage power, or the like. To connect each circuitry to the processor, another circuitry, a power source (e.g., battery) of the HMD, or any combination thereof, the HWD includes one or more flexible circuits. These flexible circuits, for example, include one or more electrically conductive layers each having one or more traces (e.g., copper traces) configured to carry power, a signal, or both. To connect a circuitry to the processor, another circuitry, a power source (e.g., battery) of the HMD, or any combination thereof, a flexible circuit is configured to be disposed on a surface of the housing (e.g., frame) of the HWD or within the housing of the HWD such that the flexible circuit is connected at a first end to the circuitry and connected at a second end to the processor, another circuitry, a power source (e.g., battery) of the HWD, or any combination thereof. For example, a flexible circuit is disposed on a surface of or within the frame of an HWD such that the flexible circuit, at a first end, is connected to a first circuitry, runs along a brow of the frame, and is connected, at a second end, to the processor. As another example, a flexible circuit is disposed on a surface of or within the frame of an HWD such that the flexible circuit, at a first end, is connected to a first circuitry, runs through the hinge of the frame, runs along at least a portion of a temple of the frame, and is connected, at a second end, to a power source. To connect a flexible circuit to the processor, another circuitry, or a power source, the flexible circuit includes one or more connectors (e.g., board-to-board connectors) each disposed at a corresponding end of a flexible circuit. These connectors, for example, are configured to connect the flexible circuit to one or more circuitries, power sources, or both by mating with one or more pins, plugs, or both of the circuitries or power sources.

[0014] However, while each of the circuitries (e.g., display circuitry, connectivity circuitry, sensing circuitry, audio circuitry) are operating, they each generate an amount of heat based on the amount of power consumed by the circuitry. That is to say, one or more components of a circuitry increase

in temperature based on the amount of power consumed by the circuitry. This increase in temperature by a circuitry, for example, causes the processor to throttle the performance of the circuitry, decrease power to the circuitry, or both so as to lower the temperature of the components. Additionally, such an increase in temperature by a circuitry causes one or more points on the housing (e.g., frame) of the HWD to increase in temperature, causing one or more hotspots on the housing of the HWD that are uncomfortable to the user. This discomfort to the user reduces the amount of time the user is able to operate the HMD and negatively impacts user experience.

[0015] As such, systems and techniques disclosed herein are directed to an HWD having one or more flexible circuits configured for heat spreading. To this end, one or more flexible circuits include a thermal tab disposed at one or both ends of the flexible circuit. These thermal tabs each include an amount of thermally conductive material, for example, copper, graphite, silver, gold, aluminum, or the like, that has a shape (e.g., bracket, square, circle) configured to thermally couple (e.g., mechanically couple so as to conduct heat generated from) to one or more circuitries. For example, a thermal tab is configured to mechanically couple to one or more heat-generating components of a circuitry such as a light engine, processor, sensor, microprocessor, or any combination thereof such that heat generated by the components is conducted by the flexible circuit. Due to the thermal tab thermally coupling a flexible circuit to a circuitry, heat generated by the circuitry is conducted by the flexible circuit, spreading the heat generated by the circuitry along the surfaces of the flexible circuit. Additionally, because the flexible circuit is disposed on a surface of or within the housing (e.g., frame) of the HWD, the heat generated by the circuitry is also spread out across the portion of the housing of the HWD along which the flexible circuit runs. For example, a flexible circuit is configured so as to spread heat conducted by a coupled circuitry along a brow or temple of the frame of the HWD along which the flexible circuit also runs. Further, to help the flexible circuit spread heat generated by a coupled circuitry, the flexible circuit includes a thermally conductive layer. This thermally conductive layer, for example, includes a layer of material (e.g., copper, graphite, silver, gold, aluminum) that runs along the length of the flexible circuit. Because the thermally conductive layer has a greater thermal conductivity than the electrically conductive layers of the flexible circuit, the amount of heat the flexible circuit is able to conduct (e.g., spread along the flexible circuit and the housing of the HMD) is increased.

[0016] In this way, the flexible circuits of the HWD are configured to spread heat generated by one or more coupled circuitries. Further, by spreading the heat generated by these circuitries, the temperature of the coupled circuitries is decreased, allowing the circuitries to increase processing speeds, power consumption, or both without overheating. Additionally, by spreading the heat generated by these circuitries, the likelihood of hotspots occurring on the housing of the HWD is reduced, helping to improve the comfort of the user and to improve the user experience.

[0017] FIG. 1 illustrates an example display system **100** housing a projection system and configured to dissipate heat using flexible circuits, in accordance with embodiments. In embodiments, the display system **100** includes a support structure **102** having an arm **104**, which houses a projection system configured to project XR images toward the eye of a user such that the user perceives the projected XR images as being displayed in a field of view (FOV) area **106** of a display at one or both of lens elements **108**, **110**. In the depicted embodiment, the display system **100** is an eyewear display that includes a support structure **102** configured to be worn on the head of a user and has a general shape and appearance of an eyeglasses (e.g., sunglasses) frame. For example, referring to the example embodiment in FIG. 1, the support structure **102** includes a first portion **120** configured to contain lens elements **108**, **110**. Additionally, this first portion **120** includes a surface that forms a brow **118** of the support structure **102**. Further, the support structure includes a first arm **104-1** terminating at a first end in a first temple **114-1** and mechanically coupled at a second, opposite end to the first portion **120** by a first hinge **116-1**. The first hinge **116-1**, for example, is configured to move so as to change the first arm **104-1** from a first position (e.g., vertical position) perpendicular to the first portion **120** to a second position (e.g., horizontal

position) parallel to the first portion **120**. In embodiments, the support structure **102** also includes a second arm **104-2** terminating at a first end in a second temple **114-2** and mechanically coupled at a second, opposite end to the first portion **120** by a second hinge **116-2**. The second hinge **116-2**, for example, is configured to move so as to change the second arm **104-2** from a first position (e.g., vertical position) perpendicular to the first portion **120** to a second position (e.g., horizontal position) parallel to the first portion **120**. Further, according to embodiments, the support structure **102** contains or otherwise includes various components of a projection system to facilitate the projection of such images toward the eye of the user, such as a display circuitry (e.g., projector, optical scanner, microdisplay, μ LED projector, liquid crystal on silicon (LCOS) projector) and a lightguide.

[0018] In embodiments, one or both of the lens elements **108**, **110** contained by support structure **102** are used by the display system **100** to provide an XR display in which rendered graphical content can be superimposed over or otherwise provided in conjunction with a real-world view as perceived by the user through the lens elements **108**, **110**. For example, light used to form a perceptible XR image or series of XR images may be projected by a projection system of the display system **100** onto the eye of the user via a series of optical path components, such as a lightguide formed at least partially in the corresponding lens element, one or more scan mirrors, one or more optical relays, one or more lenses, or any combination thereof. In embodiments, one or both of the lens elements **108**, **110** include at least a portion of a lightguide that routes display light received by an incoupler of the lightguide to an outcoupler of the lightguide, which outputs the display light toward an eye of a user of the display system **100**. The display light is imaged onto the eye of the user such that the user perceives the display light as an image. In addition, each of the lens elements **108**, **110** is sufficiently transparent to allow a user to see through the lens elements to provide a field of view of the user's real-world environment such that the XR image appears superimposed over at least a portion of the real-world environment.

[0019] In some embodiments, the projection system is a digital light processing-based projector, a microdisplay, a scanning laser projector, a LCoS projector, or any combination of a modulative light source such as a laser or one or more LEDs (e.g., μ LEDs) and a dynamic reflector mechanism such as one or more dynamic scanners or digital light processors. The projection system is communicatively coupled to the controller and a non-transitory processor-readable storage medium or memory storing processor-executable instructions and other data that, when executed by the controller, cause the controller to control the operation of the light engine. In some embodiments, the controller controls a scan area size and scan area location for the light engine and is communicatively coupled to a processor (not shown) that generates content to be displayed at the display system **100**. The projection system scans light over a variable area, designated the FOV area **106**, of the display system **100**. The scan area size corresponds to the size of the FOV area **106** and the scan area location corresponds to a region of one of the lens elements **108**, **110** at which the FOV area **106** is visible to the user. Generally, it is desirable for a display to have a wide FOV to accommodate the outcoupling of light across a wide range of angles. Herein, the range of different user eye positions that will be able to see the display is referred to as the eyebox of the display.

[0020] In embodiments, the support structure **102** contains or otherwise includes components of additional circuitries that add further functionality to the display system **100**. For example, according to embodiments, the support structure **102** contains or otherwise includes a sensing circuitry configured to take one or more measurements of the user, the HWD, the environment surrounding the user, or any combination thereof. The display system **100** then uses such measurements to track the gaze of the user, determine where a user is looking, determine objects in front of the user, determine a pose of the user, determine a pose of the display system **100**, determine a location of the user, or any combination thereof. To take such measurements, the sensing circuitry includes various sensors, such as one or more front-facing cameras, rear-facing cameras, other light sensors, motion sensors, accelerometers, radar sensors, proximity sensors,

infrared sensors, GPS sensors, and the like. As another example, in embodiments, the support structure **102** contains or otherwise includes a connectivity circuitry configured to communicatively couple the display system **100** to one or more other devices (e.g., smartphones, laptops, computers, tablets, wearable devices), networks (e.g., local area networks, wide area networks, cellular networks, the internet), or both using one or more wireless communication protocols, for example, Bluetooth protocols, cellular (e.g., 2G, 3G, 4G, 5G) protocols, WI-FI protocols, or the like. To communicatively couple the display system **100** to one or more other devices, networks, or both, the connectivity circuitry includes one or more radio frequency (RF) interfaces or other wireless interfaces, such as a Bluetooth™ interface, a WI-FI interface, and the like. As yet another example, in some embodiments, the support structure **102** contains or otherwise includes an audio circuitry configured to receive and output audio such as music, voice, noises, and the like. For example, the audio circuitry is configured to receive and output audio for one or more applications running on the display system **100** or a device connected to the display system **100** such as streaming applications, music applications, video applications, audio calling applications, video calling applications, and the like. To receive and output audio for these applications, the audio circuitry includes one or more speakers, microphones, digital signal processors, and the like,

[0021] Further, in some embodiments, the support structure contains or otherwise includes one or more processors configured to execute applications, manage the functionality of the circuitries (e.g., display circuitry, sensing circuitry, connectivity circuitry, audio circuitry), manage power, or any combination thereof. Such processors, for example, include one or more central processing units (CPUs), graphics processing units (GPUs), microcontrollers, power controllers, programmable logic devices, and the like. According to embodiments, the support structure **102** further contains or otherwise includes one or more power sources (e.g., batteries or other portable power sources) for supplying power to the circuitries (e.g., display circuitry, sensing circuitry, connectivity circuitry, audio circuitry) and processors of the display system **100**. Additionally, in embodiments, to connect each circuitry of the display system **100** to another circuitry, power source, processor, or any combination thereof, the display system **100** includes one or more flexible circuits. These flexible circuits, for example, include one or more electrically conductive layers each configured to carry a signal or power from a first end of the layer to a second end of the layer. For example, each electrically conductive layer of a flexible circuit includes a flexible material (e.g., a flexible polymer, polyimide film) having one or more traces (e.g., copper traces) configured to carry one or more signals, power, or both from a first end of the layer to a second end of the layer. To connect a circuitry to another circuitry, processor, or power source, a flexible circuit, at a first end, is first electrically coupled to the circuitry. The flexible circuit then runs along a portion of an outer surface or an interior surface of the support structure **102** and is electrically coupled, at a second end, to the other circuitry, processor, or power source. To electrically couple a flexible circuit to a circuitry, processor, or power source, a flexible circuit includes one or more connectors each disposed at a corresponding end of the flexible circuit. Each connector, for example, includes one or more pins, contacts, or both configured to electrically couple one or more layers of the flexible circuit to the pins or contacts of a mated connector (e.g., another connector mated to the connector of the flexible circuit).

[0022] In some embodiments, some or all of these components of the circuitries (e.g., display circuitry, sensing circuitry, connectivity circuitry, audio circuitry) and processors of the display system **100** are fully or partially contained within an inner volume of support structure **102**, such as within the arm **104** in region **112** of the support structure **102**. However, while a circuitry of the display system **100** is operating, one or more components of the circuitry are configured to generate an amount of heat based on the amount of power consumed by the component. As an example, while a display circuitry of the display system **100** is operating, the light engine and optical scanner both generate an amount of heat based on the amount of power consumed. As another example,

while a sensing circuitry of the display system **100** is operating one or more cameras, sensors, or both generate an amount of heat based on the amount of power consumed. As yet another example, while a connectivity circuitry of the display system **100** is operating, one or more RF or wireless interfaces generate an amount of heat based on the amount of power consumed. Due to the heat the components of these circuitries produce, the temperature of the circuitries is increased, requiring a processor of the display system **100** to throttle or reduce power to the circuitry so as to lower the temperature of the circuitry. Additionally, the heat generated by the components of these circuits causes portions of the support structure **102** to increase in temperature, causing one or more hotspots to form on the support structure **102**. These hotspots cause discomfort in the user, limiting the user's ability to use the display system **100** and negatively impacting user experience.

[0023] To this end, in embodiments, the display system **100** includes one or more flexible circuits configured to spread the heat generated by one or more components of one or more circuitries. For example, according to embodiments, one or more flexible circuits include one or more thermal tabs configured to thermally couple a flexible circuit to one or more components of a circuitry such that the flexible circuit conducts the heat generated by the components. Such thermal tabs, for example, are formed from a thermally conductive material (e.g., copper, graphite, silver, gold, aluminum) in a shape (e.g., bracket, rectangle, circle) configured to mechanically couple to one or more components of a circuitry such that the flexible circuit is thermally coupled to the components. Due to a flexible circuit being thermally coupled to the components of a circuitry, at least a portion of the heat generated by the components is conducted by the flexible circuit such that the portion of heat is spread across the surfaces (e.g., layers) of the flexible circuit. Additionally, because the flexible circuit is disposed within or on a surface of the portion of the support structure **102**, the flexible circuit is configured to spread heat from components of a circuitry to one or more portions of the support structure **102** proximate to the flexible circuit. That is to say, the flexible circuit is configured to spread heat from the components of a circuitry to the portions of the support structure **102** along with the flexible circuit runs. As an example, in embodiments, the flexible circuit running along the brow **118**, first hinge **116-1**, and first arm **104-1** of the support structure **102** is configured to spread heat generated by components of a circuitry to at least a portion of the brow **118**, first hinge **116-1**, and first arm **104-1**. As another example, according to embodiments, a flexible circuit running along the first arm **104-1** and first temple **114-1** is configured to spread heat generated by components of a circuitry to at least a portion of the first arm **104-1** and first temple **114-1**.

[0024] To help increase the amount of heat spread by a flexible circuit, in embodiments, the flexible circuit includes one or more thermally conductive layers. These thermally conductive layers, for example, include a layer of thermally conductive material (e.g., copper, graphite, silver, gold, aluminum) within or disposed on the flexible circuit. Because these thermally conductive layers have a greater thermal conductivity than the electrically conductive layers of the flexible circuit, the amount of heat able to be spread by the flexible circuit is increased. In this way, the flexible circuits of the display system **100** are configured to spread heat generated by one or more coupled components of one or more circuitries. By spreading the heat generated by the components of these circuitries, the temperature of the circuitries is decreased, allowing the circuitries to increase processing speeds, power consumption, or both without overheating. Additionally, by spreading the heat generated by these circuitries across the flexible circuits and the support structure **102**, the likelihood of hotspots occurring on the support structure **102** is reduced, helping to improve the comfort of the user and to improve the user experience.

[0025] It should be noted that while an example form factor is depicted, it will be appreciated that in other embodiments the display system **100** may have a different shape and appearance from the eyeglasses frame depicted in FIG. 1.

[0026] FIG. 2 illustrates a simplified block diagram of a projection system **200** that projects XR images directly onto the eye of a user via light, in accordance with embodiments. The projection

system **200** includes a lightguide **205** and display circuitry that includes light engine **222**, optical scanner **224**, or both. The lightguide **205** includes an incoupler **226** and an outcoupler **228**, with the outcoupler **228** being optically aligned with an eye **230** of a user in the present example. In some embodiments, the projection system **200** is implemented in an HWD, such as the display system **100** of FIG. **1**.

[0027] The light engine **222** includes one or more light sources configured to generate and output light **232** (e.g., visible laser light such as red, blue, and green laser light and/or non-visible laser light such as infrared laser light) representative of an XR image. In some embodiments, the light engine **222** is coupled to a driver or other controller (not shown), which controls the timing of emission of light from the light sources of the light engine **222** in accordance with instructions received by the controller or driver from a computer processor coupled thereto to modulate the light **232** to be perceived as images when output to the retina of an eye **230** of a user. For example, during the operation of the projection system **200**, multiple laser light beams having respectively different wavelengths are output by the light sources of the light engine **222**, then combined via a beam combiner (not shown), before being directed to the eye **230** of the user. The light engine **222** modulates the respective intensities of the light beams so that the combined light reflects a series of pixels of an XR image, with the particular intensity of each light beam at any given point in time contributing to the amount of corresponding color content and brightness in the pixel being represented by the combined light at that time.

[0028] In embodiments, the lightguide **205** of the projection system **200** includes the incoupler **226** and the outcoupler **228**. The term “lightguide,” as used herein, will be understood to mean a combiner using one or more of TIR, PIR, specialized filters, and/or reflective surfaces, to transfer light from an incoupler (such as the incoupler **226**) to an outcoupler (such as the outcoupler **228**). In general, the terms “incoupler” and “outcoupler” will be understood to refer to a set of any type of optical structures, including, but not limited to, diffraction grating structures, reflectors, mirrors, holograms, holographic optical elements (e.g., optical elements using one or more holograms), volume diffraction grating structures, volume holograms, surface relief diffraction grating structures, surface relief holograms, or any combination thereof.

[0029] In the present example presented in FIG. **2**, the display light **232** received at the incoupler **226** is relayed to the outcoupler **228** via the lightguide **205** using TIR, PIR, or both. The display light **232** is then output to the eye **230** of a user via the outcoupler **228**. In embodiments, one or more elements (e.g., incoupler **226**, outcoupler **228**, EPE) of the lightguide **205** are configured to apply an optical power to the light **232** provided from light engine **222**. For example, the incoupler **226** of the lightguide **205** includes a set of gratings (e.g., transmissive gratings, reflective gratings) configured to apply a first optical power to received light. In this way, the light **232** provided by light engine **222** has an optical power applied to it as the light **232** propagates through the lightguide **205**. As described above, in some embodiments the lightguide **205** is implemented as part of an eyeglass lens, such as the lens element **108** or lens element **110** (e.g., FIG. **1**) of the display system having an eyeglass form factor and employing the projection system **200**.

[0030] According to some embodiments, light engine **222** is configured to directly provide light **232** to the incoupler **226** of the lightguide **205** while in other embodiments, light engine **222** first provides light **232** to an optical scanner **204**. The optical scanner **204** is configured to receive light **232** and scan light **232** in one or more directions toward incoupler **226** of lightguide **205**. To this end, the optical scanner **204** includes one or more scan mirrors (e.g., MEMS mirrors) configured to scan received light in one or more directions (e.g., about one or more axes) and one or more optics relays configured to relay received light to a second point (e.g., incoupler **226**). As an example, optical scanner **204** includes one or more MEMS mirrors that are driven by respective actuation voltages to oscillate in one or more directions (e.g., about one or more axes) during active operation of the projection system **200**, causing the MEMS mirrors to scan the light **232** in one or more directions. Additionally, the optical scanner **204** includes one or more optical relays each

including lenses, reflectors, or both configured to relay scanned light from a first scan mirror to a second scan mirror, relay scanned light from a scan mirror to incoupler **226**, or both. For example, an optical relay includes a reflective relay, 2F relay, 4F relay, or any combination thereof configured to relay scanned light from a first scan mirror to a second scan mirror, incoupler **226**, or both. In embodiments, an optical relay of the optical scanner **204** includes a line-scan relay configured to, for example, receive light scanned in one or more directions from a first scan mirror and relay the scanned light to a second scan mirror, the incoupler **226**, or both such that the scanned light converges in the one or more directions to an exit pupil beyond the second scan mirror, the incoupler **226**, or both. An exit pupil, for example, refers to the location along the optical path where beams of light intersect. According to embodiments, the width (e.g., smallest dimension) of a given exit pupil approximately corresponds to the diameter of the light **232** corresponding to that exit pupil.

[0031] Although not shown in the example of FIG. 2, in some embodiments, additional optical components are included in any of the optical paths between the light engine **222** and the optical scanner **204**, between the optical scanner **204** and the incoupler **226**, between the incoupler **226** and the outcoupler **228**, between the outcoupler **228** and the eye **230** (e.g., in order to shape the display light for viewing by the eye **230** of the user), or any combination thereof. In some embodiments, a prism is used to steer display light from the optical scanner **204** into the incoupler **226** so that display light is coupled into incoupler **226** at the appropriate angle to encourage the propagation of the display light in lightguide **205** by TIR. Also, in some embodiments, an exit pupil expander (EPE), such as a set of fold grating structures, is arranged in an intermediate stage between incoupler **226** and outcoupler **228** to receive display light that is coupled into lightguide **205** by the incoupler **226**, expand the display light, and redirect the display light towards the outcoupler **228**, where the outcoupler **228** then couples the display light out of lightguide **205** (e.g., toward the eye **230** of the user). Additionally, in embodiments, one or more components of projection system **200** are thermally coupled to one or more flexible circuits such that the flexible circuits spread the heat generated by the components across the flexible circuits, a portion of a support structure **102**, or both. For example, according to some embodiments, optical scanner **224**, light engine **222**, or both are mechanically coupled to a thermal tab of a flexible circuit such that the optical scanner **224**, light engine **222**, or both are thermally coupled to a flexible circuit. The flexible circuit is then configured to spread heat generated by optical scanner **224**, light engine **222**, or both across one or more surfaces (e.g., layers) of the flexible circuit and one or more portions of a support structure **102** including or otherwise containing the optical scanner **224**, light engine **222**, or both.

[0032] Referring now to FIG. 3, an example display system **300** having one or more heat-generating components is presented, in accordance with embodiments. In embodiments, example display system **300** is implemented as at least a portion of display system **100**. According to embodiments, example display system **300** includes one or more circuitries each configured to provide a functionality to the example display system **300** and each including one or more components configured to generate heat while operating. For example, in some embodiments, example display system **300** includes a connectivity circuitry **332** configured to communicatively couple the example display system **300** to one or more devices, networks, or both such that the example display system **300** is able to transmit and receive data from the one or more devices, networks, the Internet, or any combination thereof. Such devices, for example, include smartphones, wearable devices (e.g., smart watches, smart rings, fitness trackers), computers, laptop computers, tablet computers, and the like. Additionally, such networks include, for example, local area networks, wide area networks, cellular networks (e.g., 2G networks, 3G networks, 4G networks, 5G networks), or any combination thereof. To communicatively couple the example display system **300** to one or more devices, networks, or both, connectivity circuitry includes one or more components such as interfaces **305**. These interfaces **305**, for example, include one or more RF interfaces, wireless interfaces (e.g., Bluetooth interfaces, WI-FI interfaces, cellular interfaces),

or both each having circuitry configured to communicatively couple the example display system **300** to one or more devices, networks, or both using one or more communication protocols (e.g., Bluetooth protocols, WI-FI protocols, cellular protocols). In embodiments, one or more interfaces **305** generate heat while operating. For example, one or more interfaces **305** generate an amount of heat based on the amount of power consumed by the interfaces **305**.

[0033] According to embodiments, example display system **300** further includes sensing circuitry **334** configured to take one or more measurements of the user, display system **300**, or both. As an example, sensing circuitry **334** is configured to take one or more measurements indicating the acceleration of the user or display system **300**, the direction of the gaze of the user, the presence of objects in front of the user, the location of the user or display system **300**, or any combination thereof, to name a few. Based on these measurements, for example, display system **300**, using processors **348**, is configured to determine a pose of the user of display system **300**, a location of the user or display system **300**, objects in front of the user, a gaze of the user, a focus of the user, or any combination thereof, to name a few. To take these measurements, sensing circuitry **334** includes one or more sensors **342** such as accelerometers, front-facing cameras, rear-facing cameras, infrared sensors, GPS sensors, radar sensors, proximity sensors, or the like. In embodiments, one or more sensors **342** generate heat while operating. For example, one or more sensors **342** generate an amount of heat based on the amount of power consumed by the interfaces **305**. According to some embodiments, display system **300** further includes audio circuitry **338** configured to output and receive music, sounds, voice, or any combination thereof for one or more applications running on the display system **300** (e.g., via processors **348**), a device communicatively coupled to the display system (e.g., connectivity circuitry **332**), or both. To output such music, sounds, voice, or any combination thereof, audio circuitry **338** includes one or more output devices **344** such as speakers, buzzers, alarms, digital-to-analog converters, and the like. Further, to input such music, sounds, voice, or any combination thereof, audio circuitry **338** includes one or more input devices **346** such as one or more microphones, analog-to-digital converters, or both. In embodiments, one or more output devices **344**, input devices **346**, or both generate heat while operating. For example, one or more output devices **344**, input devices **346**, or both generate an amount of heat based on the amount of power consumed by the output devices **344** and input devices **346**.

[0034] In embodiments, example display system **300** further includes display circuitry **336**. Such a display circuitry **336**, in embodiments, is implemented within projection system **200** and is configured to facilitate the presentation of an XR image to a user. To this end, display circuitry **336** includes light engine **222** and optical scanner **224**. According to embodiments, light engine **222**, optical scanner **224**, or both generate heat while operating. For example, light engine **222**, optical scanner **224**, or both generate an amount of heat based on the amount of power consumed by the light engine **222** and optical scanner **224**, respectively. As an example, as light engine **222** emits light **232** representing XR images with greater resolutions, the amount of power consumed by and the amount of heat generated by light engine **222** is increased.

[0035] To manage the operation of connectivity circuitry **332**, sensing circuitry **334**, display circuitry **336**, audio circuitry **338**, or any combination thereof, example display system **300** also includes one or more processors **348**. Each processor **348**, for example, includes a CPU, GPU, microprocessor, power controller, programmable logic device, or any combination thereof. In embodiments, one or more processors **348** are configured to run one or more applications. To execute these applications, the processors **348** send data representing instructions, operations, or both to be performed by connectivity circuitry **332**, sensing circuitry **334**, display circuitry **336**, audio circuitry **338**, or any combination thereof. To electrically couple the connectivity circuitry **332**, sensing circuitry **334**, display circuitry **336**, audio circuitry **338**, or any combination thereof to one or more processors **348**, power sources (not pictured for clarity), or both, example display system **300** includes one or more flexible circuits **340**. Each flexible circuit **340**, for example,

includes one or more flexible electrically conductive layers each configured to carry one or more signals or power from a first end of the flexible circuit **340** to a second, opposite end of the flexible circuit **340**. According to embodiments, each layer of a flexible circuit **340** is formed from a flexible material such as a flexible polymer, a polyimide film, or the like. Additionally, to enable an electrically conductive layer to carry one or more signals or power, each electrically conductive layer includes one or more traces formed from an electrically conductive material such as copper, silver, gold, and the like. These traces are each configured to provide a path for a signal or power from a first end of the flexible circuit to a second, opposite end of the flexible circuit.

[0036] Referring to the example embodiment presented in FIG. 3, example display system **300** includes a first flexible circuit **340-1** configured to electrically couple connectivity circuitry to one or more processors **348**, a second flexible circuit **340-2** configured to electrically couple sensing circuitry **334** to one or more processors **348**, a third flexible circuit **340-3** configured to electrically couple display circuitry **336** to one or more processors **348**, and a fourth flexible circuit **340-4** configured to electrically couple audio circuitry **338** to one or more processors **348**. Though the example embodiment presented in FIG. 3 shows example display system **300** as including four flexible circuits, in other embodiments, example display system **300** can include any number of flexible circuits **340**.

[0037] According to embodiments, each flexible circuit **340** is disposed on a surface of or within a support structure (e.g., support structure **102**) of the example display system **300** such that the flexible circuit **340** runs along a corresponding portion of the support structure. In this way, each flexible circuit **340** is configured to carry a signal or power from a circuitry (e.g., connectivity circuitry **332**, sensing circuitry **334**, display circuitry **336**, audio circuitry **338**) at a first location in the display system **300** (e.g., at a first location within the support structure) to one or more processors **348** or a power source at a second location in the display system **300** (e.g., at a second location within the support structure). As an example, in embodiments, a flexible circuit **340** is configured to electrically couple a circuitry in a first arm **104-1** of a support structure to a processor **348** in a second arm **104-2** of the support structure by running along at least a portion of the first arm **104-1**, a first hinge **116-1**, a brow **118**, a second hinge **116-2**, and the second arm **104-2**. As another example, a flexible circuit **340** is configured to electrically couple a circuitry in a first hinge **116-1** of a support structure to a power source in a first arm **104-1** of the support structure by running along at least a portion of the first hinge **116-1** and the first arm **104-1**. In some embodiments, one or more flexible circuits **340** are configured to help spread the heat generated by one or more circuitries (e.g., one or more components of one or more circuitries). For example, a flexible circuit **340** configured to carry a signal from a first circuitry at a first portion of a support structure to a processor **348** at a second portion of the support structure further includes a thermal tab configured to thermally couple the flexible circuit **340** to one or more components of the first circuitry. Due to the flexible circuit **340** being thermally coupled to the components of the first circuitry, the flexible circuit **340** spreads the heat generated by the components across the surfaces of the flexible circuit **340** and the portions of the support structure along which the flexible circuit **340** runs.

[0038] Referring now to FIG. 4, an example flexible circuit **400** is presented, in accordance with some embodiments. In embodiments, example flexible circuit **400** is implemented in example display system **300** as a flexible circuit **340**. According to embodiments, example flexible circuit **400** includes one or more electrically conductive layers **444**. Each electrically conductive layer **444**, for example, is formed from a flexible material (e.g., flexible polymer, polyimide film) that allows the example flexible circuit **400** to run along at least a portion of a support structure. Additionally, each electrically conductive layer **444** includes one or more traces configured to carry one or more signals or power from a first end of the example flexible circuit **400** to a second, opposite end of the example flexible circuit **400**. Such traces, for example, are formed from an electrically conductive material (e.g., copper, silver gold) and run from a first point of the

electrically conductive layer **444** to a second point of the electrically conductive layer **444**. According to embodiments, example flexible circuit **400** also includes one or more interlayers **442**. Each interlayer **442**, for example, is formed from a flexible electrically insulating material, such as a flexible polymer, polyimide film, dielectric material, prepreg, or the like. In embodiments, each interlayer **442**, is configured to adhere two electrically conductive layers **444** together. To this end, each interlayer **442** is disposed between a first electrically conductive layer **444** and a second electrically conductive layer **444** and includes an adhesive (e.g., epoxy, acrylic-based adhesive) disposed on a first surface of the interlayer **442** (e.g., to adhere the first electrically conductive layer **444**) and a second surface of the interlayer **442** (e.g., to adhere the second electrically conductive layer **444**). Though the example embodiment presented in FIG. 4 shows example flexible circuit **400** as including six electrically conductive layers **444-1**, **444-2**, **444-3**, **444-4**, **444-5**, **444-N** representing an N number of electrically conductive layers **444** and six interlayers **442-1**, **442-2**, **442-3**, **442-4**, **442-5**, **442-M** representing an M number of interlayers **442**, in other embodiments, example flexible circuit **400** can include any number of electrically conductive layers **444** and any number of interlayers **442**. In some embodiments, example flexible circuit **400** includes the same number of electrically conductive layers **444** and interlayers **442** while in other embodiments example flexible circuit **400** includes a first number of electrically conductive layers **444** and a second, different number of interlayers **442**.

[0039] According to some embodiments, example flexible circuit **400** also includes one or more vias **446**. Each via **446**, for example, is configured to electrically couple one or more traces of a first electrically conductive layer **444** to one or more traces of one or more other electrically conductive layers **444**. To this end, each via **446**, for example, is formed from an electrically conductive material such as copper, silver, gold, or the like. Referring to the example embodiment of FIG. 4, example flexible circuit **400** includes a first via **446-1** configured to electrically couple one or more traces from electrically conductive layer **444-4** to one or more traces of electrically conductive layers **444-2**, **444-3**, a second via **446-2** configured to electrically couple one or more traces from electrically conductive layer **444-5** to one or more traces of electrically conductive layers **444-1**, **444-2**, **444-3**, **444-4**, and a third via **446-3** configured to electrically couple one or more traces from electrically conductive layer **444-4** to one or more traces of electrically conductive layer **444-3**. Though the example embodiment presented in FIG. 4 shows example flexible circuit **400** as including three vias **446-1**, **446-2**, **446-3**, in other embodiments, example flexible circuit **400** can include any number of via **446**.

[0040] In embodiments, to electrically couple one or more electrically conductive layers **444** to one or more circuitries, processors **348**, power sources, or any combination thereof, example flexible circuit **400** includes one or more connectors **448**. Each connector **448**, for example, is disposed at a corresponding end (e.g., first end, second end) of example flexible circuit **400**. Further, each connector **448** includes a non-electrically conductive housing (e.g., plastic housing) containing one or more pins or contacts each electrically coupled to corresponding traces on respective electrically conductive layers **444**. To electrically couple one or more electrically conductive layers **444** to one or more circuitries, processors **348**, power sources, or any combination thereof, each connector **448** of the example flexible circuit **400** is configured to mate with a connector of the circuitry, processor **348**, or power source (e.g., a connector of the circuit board including the circuitry, processor **348**, or power source) such that the circuitry, processor **348**, or power source is electrically coupled to one or more traces of one or more electrically conductive layers **444**. However, because a connector **448** is configured to mate with another connector of a circuit, processor **348**, or power source, the connector **448** is in contact with the other connector rather than a component that generates heat, preventing the connector **448** from thermally coupling the flexible circuit **400** to the component. Additionally, in embodiments, the connector **448** includes a non-thermally conductive material, such as plastic, further preventing the connector **448** from thermally coupling the flexible circuit **400** to a component that generates heat. To this end, according to embodiments, example

flexible circuit **400** includes one or more thermal tabs.

[0041] For example, referring now to FIG. 5, a heat-spreading configuration **500** for a flexible circuit is presented, in accordance with embodiments. In embodiments, heat-spreading configuration **500** is implemented in example flexible circuit **400**. According to embodiments, heat-spreading configuration **500** includes example flexible circuit **400** including one or more thermal tabs. For example, in the embodiment presented in FIG. 5, example flexible circuit **400** includes a first thermal tab **550-1** disposed at a first end of the example flexible circuit **400** and a second thermal tab **550-2** disposed at a second, opposite end of the example flexible circuit **400**. Each thermal tab **500**, for example, is thermally coupled to one or more layers (e.g., electrically conductive layers **444**, interlayers **442**) of the example flexible circuit **400**. That is to say, each thermal tab is configured to transfer heat to one or more layers (e.g., electrically conductive layers **444**, interlayers **442**) of the example flexible circuit **400**. Further, each thermal tab **550** is formed from a thermally conductive material such as copper, graphite, silver, gold, aluminum, or the like and has a shape (e.g., bracket, rectangle, circle) mechanically coupleable to one or more components (e.g., interfaces **305**, sensors **342**, light engine **222**, optical scanner **224**, output devices **344**, input devices **346**) of one or more circuitries (e.g., connectivity circuitry **332**, sensing circuitry **334**, display circuitry **336**, audio circuitry **338**). For example, a thermal tab **500** is configured to be mechanically coupled to a component of a circuitry such that the example flexible circuit **400** and the component are thermally coupled. That is to say, a thermal tab **500** is configured to be mechanically coupled to a component of a circuitry such that heat generated by the component is spread across one or more layers of the example flexible circuit **400**. Though the example embodiment presents the heat-spreading configuration **500** as including two thermal tabs **550-1**, **550-2** each disposed at a corresponding end of the example flexible circuit **400**, in other embodiments, the heat-spreading configuration **500** includes a different number of thermal tabs **500** each disposed at corresponding points of the example flexible circuit **400**.

[0042] In some embodiments, example flexible circuit **400** is connected to one or more thermal tabs **550** by a thermal strap **552**. Such a thermal strap **552**, for example, includes a length of thermally conductive material (e.g., copper, graphite, silver gold) connected at a first end to a thermal tab **550** and a second, opposite end to one or more layers of example flexible circuit **400**. According to embodiments, thermal strap **552** includes one or more strands of thermally conductive material arranged, for example, in a braid. In embodiments, each thermal strap **552** is configured to thermally couple one or more layers of the example flexible circuit **400** to a thermal tab **550**. That is to say, thermal strap **552** is configured to transfer heat to and from the thermal tab **550** to one or more layers of the example flexible circuit **400**. Additionally, according to some embodiments, example flexible circuit **400** further includes one or more thermal layers **554**. These thermal layers **554**, for example, are each disposed on a respective surface (e.g., top surface, bottom surface) of the example flexible circuit **400** and are each formed from a thermally conductive material such as copper, graphite, silver, gold, or the like. As an example, in some embodiments, a thermal layer **554** includes a graphite sheet. In embodiments, a thermal layer **554** is configured to spread heat from one or more layers of the example flexible circuit **400**, a thermal strap **552**, a thermal tab **550**, or any combination thereof. For example, referring to the embodiment presented in FIG. 5, based on thermal tab **550-1** being mechanically coupled to a component of circuitry, at least a portion of the heat generated by the component travels through the thermal tab **550-1** and through thermal strap **552** to one or more layers (e.g., electrically conductive layers **444**, interlayers **442**) of the example flexible circuit **400** and thermal layer **554**. In some embodiments, one or more thermal tabs **550** are connected only to one or more thermal layers **554** (e.g., via a thermal strap **552**) rather than any layers of the example flexible circuit **400**. Though the example embodiment in FIG. 5 presents the heat-spreading configuration **500** as including a single thermal layer **554** disposed on a first surface (e.g., bottom surface) of the example flexible circuit **400**, in other examples, the heat-spreading configuration **500** also includes a second thermal layer **554** disposed on a second surface (e.g., top

surface) of the example flexible circuit **400**.

[0043] According to some embodiments, a thermal layer **554** includes a prefabricated assembly that is adhered to a surface (e.g., top surface, bottom surface) of the example flexible circuit **400**. For example, in some embodiments, a thermal layer **554** includes a prefabricated assembly that is adhered to a surface of the example flexible circuit **400** via a pressure-sensitive adhesive.

According to embodiments, this prefabricated assembly of the thermal layer **554** includes one or more thermal tabs **550**, thermal straps **552**, or both. In this way, during assembly of a display system, the prefabricated assembly of the thermal layer **554** is able to adhere to a surface of example flexible circuit **400** so as to implement heat-spreading configuration **500**. Because heat-spreading configuration **500** is able to be implemented during assembly by adhering the prefabricated assembly of thermal layer **554** to a surface of the example flexible circuit **400**, heat-spreading configuration **500** is able to be implemented for example flexible circuit **400** without altering or refabricating any of the other circuit boards used to assemble the display system. As such, a display system is enabled to dissipate heat using flexible circuits **400** without needing to refabricate the circuit boards of the components that generate heat.

[0044] Referring now to FIG. **6**, an example display system **600** including a flexible circuit configured for heat dissipation is presented, in accordance with embodiments. According to embodiments, example display system **600** includes a heat-generating component **656** disposed with a support structure **102**. For example, within display system **600**, the heat-generating component **656** is disposed within a hinge **116** of the support structure. Such a heat-generating component **656**, for example, includes a component of a circuitry (e.g., connectivity circuitry **332**, sensing circuitry **334**, display circuitry **336**, audio circuitry **338**) such as an interface **305**, sensor **342**, light engine **222**, optical scanner **224**, output device **344**, input device **346**, or any combination thereof. Further, according to embodiments, example display system **600** includes a thermal tab **550** mechanically coupled to the heat-generating component **656**. Such a thermal tab **550**, for example, is configured to thermally couple the heat-generating component **656** to at least a portion of an example flexible circuit **400** such as, for example, a thermal layer **554** of the example flexible circuit **400**. As an example, the thermal tab **550** is configured to thermally couple the heat-generating component **656** to at least a portion of the example flexible circuit **400** via a thermal strap **552**.

[0045] Within example display system **600**, the example flexible circuit **400** is configured to be disposed on a surface of the support structure **102** such that the example flexible circuit **400** runs along a brow **118** of the support structure **102**. Due to the example flexible circuit **400** being thermally coupled to the heat-generating component **656** via thermal tab **550**, at least a portion of the heat generated by heat-generating component **656** travel through the thermal tab **550** to the flexible circuit **400**. The flexible circuit **400** then spreads the heat over one or more surfaces of the flexible circuit **400** and at least a portion of the brow **118**. Because the flexible circuit **400** spreads at least a portion of the heat generated by heat-generating component **656** in this way, the temperature of the heat-generating component **656** is reduced, allowing the heat-generating component **656** to use more power, processor more quickly, or both. Additionally, because the flexible circuit **400** spreads at least a portion of the heat generated by heat-generating component **656** in this way, a hotspot is less likely to form on the portion of the support structure **102** proximate to the heat-generating component **656**, helping to improve the comfort of the user.

[0046] Referring now to FIG. **7**, a partially transparent view of a HWD **700** including flexible circuits configured for heat dissipation is presented, in accordance with some embodiments. In embodiments, HWD **700** is configured to provide light **218** representative of an XR image to an eye **230** of the user such that the XR image is presented to the user in a real-world setting. To this end, HWD **700** includes a display system such as display system **100**, display system **300**, or display system **600**. Further, HWD **700** includes a support structure **102** having a brow **118**, hinge **116**, arm **104**, and temple **114**. According to some embodiments, HWD **700** includes a first heat-

generating component **756** contained (e.g., housed in) the arm **104** of the support structure **102**. Such a first heat-generating component **756** includes, for example, an interface **305**, sensor **342**, light engine **222**, optical scanner **224**, output device **344**, input device **346**, or any combination thereof. The first heat-generating component **756**, for example, is thermally coupled, via a thermal tab **550**, thermal strap **552**, or both, to a first flexible circuit **760**, similar to or the same example flexible circuit **400**. The first flexible circuit **760** is disposed on a surface of or within the support structure **102** such that the first flexible circuit **760** runs along at least a portion of the arm **104**, hinge **116**, and brow **118** of the support structure **102**. Due to the first flexible circuit **760** being thermally coupled to the first heat-generating component **756**, one or more surfaces of the first flexible circuit **760**, a thermal layer **554** of the first flexible circuit **760**, or both are configured to conduct at least a portion of the heat generated by the first heat-generating component **756** such that at least a portion of the heat is spread across one or more surfaces of the first flexible circuit **760**, a thermal layer **554** of the first flexible circuit **760**, at least a portion of the arm **104**, at least a portion of the hinge **116**, and at least a portion of the brow **118**.

[0047] Further, in embodiments, HWD **700** includes a second heat-generating component **758** contained (e.g., housed in) the arm **104** of the support structure **102**. Such a second heat-generating component **758** includes, for example, an interface **305**, sensor **342**, light engine **222**, optical scanner **224**, output device **344**, input device **346**, or any combination thereof. The second heat-generating component **758**, as an example, is thermally coupled, via a thermal tab **550**, thermal strap **552**, or both, to a second flexible circuit **762**, similar to or the same example flexible circuit **400**. The second flexible circuit **762** is disposed on a surface of or within the support structure **102** such that the first flexible circuit **760** runs along at least a portion of the arm **104** and temple **114** of the support structure **102**. Due to the second flexible circuit **762** being thermally coupled to the second heat-generating component **758**, one or more surfaces of the second flexible circuit **762**, a thermal layer **554** of the second flexible circuit **762**, or both are configured to conduct at least a portion of the heat generated by the second heat-generating component **758** such that at least a portion of the heat is spread across one or more surfaces of the second flexible circuit **762**, a thermal layer **554** of the second flexible circuit **762**, at least a portion of the arm **104**, and at least a portion of the temple **114**.

[0048] In some embodiments, certain aspects of the techniques described above may be implemented by one or more processors of a processing system executing software. The software comprises one or more sets of executable instructions stored or otherwise tangibly embodied on a non-transitory computer readable storage medium. The software can include the instructions and certain data that, when executed by the one or more processors, manipulate the one or more processors to perform one or more aspects of the techniques described above. The non-transitory computer readable storage medium can include, for example, a magnetic or optical disk storage device, solid state storage devices such as Flash memory, a cache, random access memory (RAM) or other non-volatile memory device or devices, and the like. The executable instructions stored on the non-transitory computer readable storage medium may be in source code, assembly language code, object code, or other instruction format that is interpreted or otherwise executable by one or more processors.

[0049] A computer readable storage medium may include any storage medium, or combination of storage media, accessible by a computer system during use to provide instructions and/or data to the computer system. Such storage media can include, but is not limited to, optical media (e.g., compact disc (CD), digital versatile disc (DVD), Blu-Ray disc), magnetic media (e.g., floppy disc, magnetic tape, or magnetic hard drive), volatile memory (e.g., random access memory (RAM) or cache), non-volatile memory (e.g., read-only memory (ROM) or Flash memory), or microelectromechanical systems (MEMS)-based storage media. The computer readable storage medium may be embedded in the computing system (e.g., system RAM or ROM), fixedly attached to the computing system (e.g., a magnetic hard drive), removably attached to the computing system

(e.g., an optical disc or Universal Serial Bus (USB)-based Flash memory) or coupled to the computer system via a wired or wireless network (e.g., network accessible storage (NAS)).

[0050] Note that not all of the activities or elements described above in the general description are required, that a portion of a specific activity or device may not be required, and that one or more further activities may be performed, or elements included, in addition to those described. Still further, the order in which activities are listed is not necessarily the order in which they are performed. Also, the concepts have been described with reference to specific embodiments. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present disclosure.

[0051] Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any feature(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature of any or all the claims. Moreover, the particular embodiments disclosed above are illustrative only, as the disclosed subject matter may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. No limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular embodiments disclosed above may be altered or modified and all such variations are considered within the scope of the disclosed subject matter. Accordingly, the protection sought herein is as set forth in the claims below.

Claims

1. A head-worn display (HWD), comprising: a support structure; a light engine configured to emit light representative of an image; a lightguide to direct the light representative of the image to an eye of a user; a circuitry configured to provide a first functionality; and a flexible circuit configured to: electrically couple the circuitry to a processor or a power source; and thermally couple to the circuitry such that heat generated by the circuitry is spread across a surface of the flexible circuit.
2. The HWD of claim 1, wherein the flexible circuit includes a thermal tab configured to thermally couple the circuitry to the flexible circuit.
3. The HWD of claim 1, wherein the flexible circuit includes a thermal layer disposed on the surface of the flexible circuitry and configured to spread heat from the circuitry.
4. The HWD of claim 1, wherein the flexible circuit is configured to run along an arm, hinge, or brow of the support structure.
5. The HWD of claim 1, wherein the flexible circuit includes a plurality of electrically conductive layers each configured to carry a respective signal.
6. The HWD of claim 1, wherein the flexible circuit is configured to spread at least a portion of the generated by the circuitry across at least a portion of the support structure.
7. The HWD of claim 1, wherein the circuitry comprises at least one from a group of a connectivity circuitry, sensing circuitry, display circuitry, and audio circuitry.
8. A flexible circuit for a head-worn display (HWD), comprising: a flexible electrically conductive layer configured to electrically couple a circuitry of the HWD to a processor or a power source of the HWD; and a thermal tab configured to thermally couple the circuitry to the flexible circuit such that heat generated by the circuitry is spread across a surface of the flexible circuit, wherein the flexible circuit is configured to run along at least a portion of a support structure of the HWD.
9. The flexible circuit of claim 8, further comprising: a thermal layer adhered to the surface of the flexible circuit, the thermal layer configured to spread at least a portion of the heat from the

circuitry across the thermal layer.

10. The flexible circuit of claim 9, further comprising: a thermal strap configured to thermally couple the thermal tab to the thermal layer.

11. The flexible circuit of claim 8, wherein the flexible circuit is configured to run along an arm, hinge, or brow of the support structure of the HWD.

12. The flexible circuit of claim 8, wherein the flexible circuit is configured to spread at least a portion of the heat generated by the circuitry across the at least a portion of the support structure.

13. The flexible circuit of claim 8, further comprising: a second flexible electrically conductive layer; and an interlayer disposed between the flexible electrically conductive layer and the second flexible electrically conductive layer.

14. The flexible circuit of claim 8, wherein the thermal tab is configured to mechanically couple to a component of the circuitry.

15. The flexible circuit of claim 8, wherein the circuitry comprises at least one from a group of a connectivity circuitry, sensing circuitry, display circuitry, and audio circuitry.

16. A method, comprising: electrically coupling, within a head-worn display (HWD) and by a flexible circuit, a circuitry of the HWD to a processor or power source of the HWD; and thermally coupling the circuitry to the flexible circuit such that heat generated by the circuitry is spread across a surface of the flexible circuit and at least a portion of a support structure of the HWD.

17. The method of claim 16, wherein the flexible circuit includes a thermal layer adhered to a surface of the flexible circuit, the thermal layer configured to spread at least a portion of the heat from the circuitry across the thermal layer.

18. The method of claim 16, wherein the flexible circuit is configured to run along an arm, hinge, and brow of the support structure of the HWD.

19. The method of claim 16, wherein the flexible circuit includes a plurality of electrically conductive layers.

20. The method of claim 16, wherein the circuitry comprises at least one from a group of a connectivity circuitry, sensing circuitry, display circuitry, and audio circuitry.
